

The SIDIS calculation through NLO

Six channels with reverse unitarity, Kira and SubTropica

Calculation-method report for the local SIDIS workflow

7 September 2026

This report explains the accepted calculation in `scripts/SIDIS/`. The outputs are the symbolic partonic structure functions $\hat{F}_{1,ij}$ and $\hat{F}_{2,ij}$ for all six channels. The notation and definition of the observable follow the supplied SIDIS paper [1]. The real integral calculation follows the reverse-unitarity approach of the supplied standalone reference [2]. Equations below state the definitions used by those sources and the programs [4], or quote saved tool results. No new physics calculation was performed to write this report.

The implemented routes are

real: FeynArts $\rightarrow$ FeynCalc $\rightarrow$ reverse unitarity $\rightarrow$ Kira $\rightarrow$ SubTropica,
virtual and self energies: FeynArts $\rightarrow$ FeynCalc $\rightarrow$ Kira $\rightarrow$ SubTropica.

These are followed by renormalization where needed, collinear factorization, pole cancellation and the final structure-function projection. The unchanged generated amplitudes, unintegrated tensors, Born tensors and observable definitions are imported from the existing channel calculations. The required real, virtual and self-energy master values are evaluated in the new integral workflow. Package-X/PaVe values and the paper's evaluated angular-integral tables do not supply those values.

How to read this report. The common sections define the observable, amplitude algebra, real and virtual integration, and subtraction. Each channel then has its own section stating its generated process, Born input where present, flavor weights, subtraction routes and final output. The numerical sections follow an $\hat{F}$ into a measured transverse-momentum bin: kinematic maps, PDFs, fragmentation, acceptance, endpoint subtraction, integration, and the two saved figures. The final file tables identify the full symbolic coefficients, whose long expressions remain in executable Wolfram Language form.

Scope of the calculation. The hard scattering is massless, unpolarized, and evaluated through NLO at nonzero transverse momentum. Symbolic N_c , N_f and quark charges are retained in the production hard parts. Numerical choices such as $N_c = 3$, active flavors, a common scale and neutral-pion fragmentation are made by the numerical consumer. This document describes the accepted run and its stated limitations; it does not replace the calculation by a derivation performed while writing the report.

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1 Observable, momenta and channels

The process is

$$\ell(l) + H(P_H) \longrightarrow \ell'(l') + h(P_h) + X, \quad \gamma^*(q) + i(p) \longrightarrow j(k_1) + X. \quad (1)$$

Here P_H denotes the incoming hadron momentum, and P_h the observed hadron momentum. The first channel label is the incoming parton; the second is the observed, fragmenting parton. Its momentum is always k_1 . Only the unobserved partons are integrated in the partonic hard tensor. The calculation is unpolarized and uses the azimuth-independent F_1, F_2 decomposition of the paper.

The defining relations are

$$D = 4 - 2\epsilon, \quad q^2 = -Q^2 < 0, \quad p^2 = k_1^2 = k_2^2 = k_3^2 = 0, \quad (2)$$

$$s = (p + q)^2, \quad t = (q - k_1)^2, \quad u = (p - k_1)^2, \quad (3)$$

$$P_X = p + q - k_1, \quad w = P_X^2 = s_{23}, \quad s + t + u = -Q^2 + w, \quad (4)$$

$$\hat{x} = \frac{Q^2}{2p \cdot q} = \frac{x}{\xi}, \quad \hat{z} = \frac{p \cdot k_1}{p \cdot q} = \frac{z}{\zeta}, \quad p = \xi P_H, \quad k_1 = P_h / \zeta. \quad (5)$$

These are the paper's definitions, implemented and checked in shared S02. The Born and virtual kinematics have $w = 0$. The interior real-emission domain recorded in the final files is

$$Q^2 > 0, \quad s > w > 0, \quad w - Q^2 - s < t < -\frac{Q^2 w}{s}. \quad (6)$$

Endpoint distributions are defined separately at $w = 0$.

1.1 What is observed and what is integrated

There are two different integrations. At the parton level, p, q, k_1 are external data, and the unresolved final state is integrated to construct $\widehat{W}^{\mu\nu}$. At the hadron level, the resulting hard functions are convolved with the incoming PDF and the fragmentation function of the observed parton. Reverse unitarity is used for the first integration. The PDF/FF convolution and the experimental bin integration are performed later.

The observed-particle label matters even when two rows contain the same species. Hqq and Hqg share the Born scattering $\gamma^* q \rightarrow qg$, but attach fragmentation to different outgoing legs. Their definition of t , the tensor contractions, the fragmentation subtraction and the final coefficient must all use their own k_1 assignment.

1.2 Perturbative order and the photon convention

At nonzero transverse momentum the leading hard parts are of order α_s , and this calculation keeps the correction of order α_s^2 . This is the meaning of LO and NLO in this report. The following table lists the generated processes; an underline marks k_1 . The photon is understood on the incoming side of every row.

Folder	Incoming	Born final state	Real final states at α_s^2
Hqq	q_i	<u>q_i</u> g	<u>q_i</u> gg ; <u>q_i</u> $q_i \bar{q}_i$; <u>q_i</u> $q_j \bar{q}_j$ with $j \neq i$
Hqg	q_i	<u>g</u> q_i	<u>g</u> $q_i g$
Hgq	g	<u>q_i</u> $\bar{q}_i$	<u>q_i</u> $\bar{q}_i g$
Hgg	g	absent	<u>g</u> $q_f \bar{q}_f$
Hqqbar	q_i	absent	<u>$\bar{q}_i$</u> $q_i q_i$
Hqqprime	q_i	absent	<u>q_j</u> $q_i \bar{q}_j$, $j \neq i$

Hqq, Hqg and Hgq require one-loop corrections to their Born processes. Hgg, Hqqbar and Hqqprime first contribute at the real-emission order shown; they have no virtual contribution at this order.

The electromagnetic coupling used in the lepton cross section is kept outside the partonic F hats. Squared quark charges remain in the hard functions with the conventions stated for each channel. The generated Born, real and one-loop amplitudes carry the recorded coupling degrees (e, g_s) of $(1, 1)$, $(1, 2)$ and $(1, 3)$, respectively. The virtual contribution is the interference with the Born amplitude. The square of a one-loop amplitude is outside this NLO calculation.

Symbols in the saved files. The symbol Q2 means Q^2 , w means s_{23} , and SUNN or Nc means N_c . The symbols mu and mu2 retain their source-specific scale conventions; the latter is a squared scale. In the numerical sections, $p_T = |\mathbf{P}_{hT}|$ is the hadron transverse momentum and z is the hadron fragmentation variable of the paper.

2 From diagrams to two scalar contractions

2.1 Generation and the reusable inputs

FeynArts supplies the diagrams and amplitudes. FeynCalc performs the spin and color algebra and the tensor contractions. The imported inputs retain the generated diagram factors, incoming averages, photon-charge removal, observed-particle tags and identical-particle bookkeeping. The imported real interference pieces must sum exactly to the saved unintegrated tensors. In the virtual route the Born interference must reproduce each channel's own accepted Born contractions.

The generation programs use the massless QCD part of the FeynArts **SMQCD** model. Requested external fields, loop order and observed-leg ordering define each process. The core scattering sources exclude tadpoles and external wave-function corrections from diagram generation; external-field effects enter at renormalization. Internal fields are restricted to the intended QCD content. FeynCalc's **FCFAConvert** converts the amplitudes to D dimensions, with the recorded mass substitutions and coupling conventions. Original diagrams and amplitude lists are preserved alongside the scalar inputs.

These inputs remain useful when the integration method changes. The new import accepts the generated amplitudes, Born contractions, unintegrated real tensors, virtual integrands and loop-independent subtraction definitions, with their recorded identities. An evaluated old real angular integral or an old Package-X virtual value is not used as a new master integral.

2.2 Interferences, spin sums and color algebra

The square is assembled from generated diagram pairs. Diagonal pairs and off-diagonal interferences carry the factors recorded by the pair program. FeynCalc performs fermion spin sums, Dirac traces, polarization sums and $SU(N_c)$ color contractions before scalar integral reduction. Pair results can be cached separately, so the complete unsimplified diagram sum need not be held in memory at once.

The saved color and spin measurements in the core Born records give

$$C_F = \frac{(N_c - 1)(N_c + 1)}{2N_c}, \quad T_F = \frac{1}{2}, \quad C_A = N_c, \quad (7)$$

$$n_{\text{spin},q} = 2, \quad n_{\text{spin},g} = D - 2, \quad (8)$$

$$A_q = \frac{1}{2N_c}, \quad A_g = \frac{1}{(D-2)(N_c-1)(N_c+1)}. \quad (9)$$

These equations quote the stored color traces and physical spin projectors. The incoming-gluon average remains D dimensional. Its dependence on the regulator must be retained in pole-multiplied terms.

Physical gluon polarization sums use the reference vectors prescribed by the source, with p and k_1 supplying the relevant lightlike directions. Photon and gluon Ward tests are evaluated on the appropriate generated amplitudes or contractions. A Ward test checks gauge consistency; it does not by itself fix a spectator factor, an incoming average or the normalization of a separate program.

2.3 Charges and identical particles

The source model has definite quark charges. Core normalization programs measure the model photon vertex, remove the model charge and expose the physical charge symbols used by the channel. In four-quark sectors, the two photon-charge assignments are retained through their interference. Flavor summation is performed at the documented assembly boundary.

The program distinguishes the observed tag from unobserved identical particles. Core tagged squares obtain their weight from the generated final-species list. The real-only convention leaves the unobserved spectator factor outside the imported tensor. Each channel section states which weight is already present and which enters at final assembly.

2.4 Tensor projection and dimensional dependence

The two contractions are

$$T_g = g^{\mu\nu} \widehat{W}_{\mu\nu}, \quad T_{pp} = p^\mu p^\nu \widehat{W}_{\mu\nu}. \quad (10)$$

The first uses $g^{\mu\nu}$ with the sign stated here. A sign change to a physical-photon polarization sum must not be inserted into this definition. Each charge and real-emission sector is retained separately until its required weights have been applied.

The paper's partonic decomposition is

$$\widehat{W}_{ij}^{\mu\nu} = \left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2} \right) \widehat{F}_{1,ij} + \frac{\widehat{p}^\mu \widehat{p}^\nu}{p \cdot q} \widehat{F}_{2,ij}, \quad \widehat{p}^\mu = p^\mu - q^\mu \frac{p \cdot q}{q^2}. \quad (11)$$

The saved D -dimensional projector definitions are

$$\mathcal{P}_1^{(D)}[T_g, T_{pp}] = \frac{-T_g/2 + 2\hat{x}^2 T_{pp}/Q^2}{1 - \epsilon}, \quad (12)$$

$$\mathcal{P}_2^{(D)}[T_g, T_{pp}] = \frac{-\hat{x} T_g + 4(3 - 2\epsilon)\hat{x}^3 T_{pp}/Q^2}{1 - \epsilon}. \quad (13)$$

The Born terms used in subtraction retain their D dependence. The final projection uses the finite combined contractions only after their poles have canceled.

The projector program constructs the decomposition, contracts it with $g^{\mu\nu}$ and $p^\mu p^\nu$, and solves the linear system. It checks the resulting projectors against the defining tensor and the paper. Thus projectors and Born tensors share the same sign and dimension convention. In particular, Pg means the $g^{\mu\nu}$ contraction; the source explicitly accounts for its sign relative to the virtual-photon polarization-sum operation.

Keeping both scalar contractions through real, virtual and collinear assembly lets the pole test use one normalization. The final projector acts on finite contractions. The Born tensor inside

subtraction remains D dimensional because poles multiplying its regulator dependence contribute to the finite result.

3 Real integration by reverse unitarity

3.1 The two physical cuts

For real emission the common loop routing is

$$k_2 = r, \quad k_3 = P_X - r. \quad (14)$$

The two on-shell constraints become the two cut propagators of one loop. In discontinuity notation the positive-energy cut is

$$\left[\frac{1}{k^2} \right]_{\text{cut},+} \equiv \frac{\theta(k^0)}{2\pi i} \left(\frac{1}{k^2 - i0} - \frac{1}{k^2 + i0} \right) = \delta_+(k^2), \quad \delta_+(k^2) = \theta(k^0)\delta(k^2). \quad (15)$$

The energy restriction is part of the physical master definition. It is retained when evaluating the cut masters, in addition to marking the cuts in the Kira reduction.

The master measure used in SIDIS is explicitly

$$M[\mathcal{R}] = \int d^D r \delta_+(r^2) \delta_+((P_X - r)^2) \mathcal{R}(r). \quad (16)$$

There are no absorbed 2π factors in this definition. The observed k_1 is fixed and is not another integrated cut. Thus a tree-level real contribution becomes a cut one-loop integral for its unresolved phase space. It does not become a virtual amplitude with an unrestricted loop.

The conversion back to the paper's unobserved-state phase space uses the normalizations already present in its Eqs. (34) and (38):

$$N_{\text{ps}}(D) = (2\pi)^D \prod_{a=2,3} (2\pi)^{-(D-1)}, \quad N_H = (2\pi)^{-4}. \quad (17)$$

The program derives this product and checks it against each channel's preserved phase-space definition. Dimensional scale factors, charge and strong-coupling factors, and any remaining spectator weights are inserted at the documented assembly boundary. They are not part of Eq. (16).

3.2 Scalar families and Kira reduction

FeynCalc expresses the contractions as rational loop integrands. Shared S02 defines candidate families from the actual tree propagator routings, including r^2 , $(P_X - r)^2$ and independent uncut denominators. Shared S03 maps every interference to these families and reconstructs its original integrand exactly.

Algebraic elimination of dependent denominators is used to construct valid families. That algebra does not evaluate a phase-space integral. The integration reduction is performed by Kira, using integration-by-parts (IBP) relations. Their usual schematic form is

$$0 = \int d^D r \left[\frac{\partial}{\partial r^\mu} (v^\mu \mathcal{I}(r)) \right]_{\text{cut}}, \quad v \in \{r, p, q, k_1\}, \quad (18)$$

with the cut-sector rules imposed on the resulting integrals. Both physical cuts must remain in every surviving master. A pinched cut sector cannot be treated as an ordinary uncut contribution.

The common scalar coordinates. The family construction uses the basis $\{p, q, k_1, r\}$ and solves the defining scalar-product equations with coordinates

$$z_1 = r^2, \quad z_2 = (P_X - r)^2, \quad a = (p - r)^2, \quad b = (k_1 + r)^2. \quad (19)$$

The candidate uncut momentum list is

$$\{p - r, p - (P_X - r), k_1 + r, k_1 + (P_X - r), q - r, q - (P_X - r)\}. \quad (20)$$

On the cuts these quadratic denominators are affine in a, b . The code chooses independent pairs by their determinant and checks independence again in the full (z_1, z_2, a, b) coordinates. The accepted geometry has twelve candidate families. Each starts with the two physical cuts, followed by an independent pair of uncut propagators. Every inherited scalar-product map is checked against its defining equations in the new coordinates.

Targets and reduction bounds. The mapper records integral-index tuples for every numerator and denominator term. Positive indices specify denominator powers; nonpositive uncut indices specify absent denominators or numerators. Kira receives the actual targets, their sectors and preferred masters retaining both cuts. Reduction bounds are measured from these targets: the r bound is the largest positive-index sum plus the number of loop momenta, and the s bound is the largest numerator-index sum. The run therefore requests the domain needed by the current integrands.

Kira performs initialization, triangular reduction and back substitution, then exports exact rules. The importer requires every target to have a rule or be a declared master, unique rule left-hand sides, and complete termination in the final master inventory. Both physical cuts must have positive indices in every master. Kira and Fermat operate on symbolic dimension and invariants throughout this production reduction.

The result is a rational linear combination

$$R_{c,\Gamma}^{\text{cut}}(\epsilon; s, t, Q^2, w) = \sum_a c_{c,\Gamma a}(D; s, t, Q^2, w) M_a(D; s, t, Q^2, w), \quad (21)$$

where c labels a channel and charge sector, and $\Gamma = g, pp$. The accepted shared reduction has **315 requested real integrals and eight cut masters**. These counts are reported by the saved Kira run; the same master set serves the six channels, with different coefficients.

3.3 What SubTropica evaluates

Kira supplies relations between integrals. SubTropica evaluates the remaining master integrals from explicit parameter representations. In this workflow it is used as an integration engine, with HyperIntica as its backend; a master name alone is not treated as a table lookup. Wolfram Language still coordinates the symbolic work and hosts these packages.

Shared S08 solves the two cut equations in the P_X rest frame, selects the positive-energy solution and derives the Jacobian. The saved tool results are

$$r^0 = |\mathbf{r}| = \frac{\sqrt{w}}{2}, \quad J_{\text{cut}} = 2^{-3+2\epsilon} w^{-\epsilon}, \quad \Omega(n) = \frac{2\pi^{n/2}}{\Gamma(n/2)}. \quad (22)$$

Here J_{cut} is the radial measure after imposing both cuts, and $\Omega(n)$ is the surface area of the unit sphere in n Euclidean dimensions. For example, the actual volume-master input is represented as

$$M_0 = J_{\text{cut}} \Omega(D-2) \int_0^\infty dv [1 - c(v)^2]^{-\epsilon} \left| \frac{dc(v)}{dv} \right|, \quad c(v) = \frac{1-v}{1+v}. \quad (23)$$

This is an input to SubTropica, not an inserted evaluated angular table.

On the cuts, every remaining propagator is reconstructed as a linear scalar product with an effective external vector. The resulting eight masters fall into four evaluated Euler classes:

Saved class	Parameter integral supplied to SubTropica
<code>cut_volume</code>	The positive-energy two-particle cut volume.
<code>cut_single_massive</code>	One uncut denominator with a non-null effective vector.
<code>cut_pair_one_massive</code>	Two denominators, with one null effective vector.
<code>cut_pair_massless</code>	Two null effective vectors; a separate input before expansion.

The pair inputs use the unexpanded Euler representation referenced in shared S08 [3]. Its parameters are obtained from the current master geometry. The program retains the radial and Gamma-function prefactors separately and calls **STIntegrate** on an ordinary parameter measure. The tuple interface supplies its own internal conversion to a logarithmic measure; the adapter must not insert that factor twice.

Raw engine output, algebraic-root definitions, physical substitutions and required ϵ orders are saved. The volume class is evaluated through ϵ^2 , and the other production cut classes through ϵ^1 in this run. The required orders are checked against the reduction coefficients and endpoint needs. An independent normalization check compares two different representations of the same null-denominator cut integral.

The accepted real route is shared S08/S10/S15 together with the S04 Kira reduction. The earlier shared S06/S07 uncut-parent route is retained as history and is not the source of the accepted physical cut-master values.

3.4 Explicit Euler inputs and their geometric parameters

This subsection records the defining parameter integrals constructed in shared S08. Let v_i be the effective vector reconstructed for an uncut denominator on the two-cut surface. The program verifies that denominator against $2r \cdot v_i$ and defines

$$S_i = P_X \cdot v_i, \quad m_i^2 = \frac{w v_i^2}{S_i^2}, \quad \beta_i = \sqrt{1 - m_i^2}. \quad (24)$$

Here m_i^2 is a dimensionless geometric parameter, not a physical particle mass. Effective vectors, their powers and all substitutions are saved for each Kira master. This is how the common classes acquire the actual s, t, Q^2, w dependence of a channel calculation.

For the single-denominator class the defining density is

$$M_{\text{single}} = \frac{J_{\text{cut}} \Omega(D-2)}{S_1} \int_0^\infty dv \frac{[1 - c(v)^2]^{-\epsilon} |c'(v)|}{1 - \beta_1 c(v)}, \quad c(v) = \frac{1 - v}{1 + v}. \quad (25)$$

The tuple uses $z_* = (1 - \beta_1)/(1 + \beta_1)$ as its algebraic parameter. SubTropica evaluates the parameter integral; the regulated radial and sphere factors multiply its output afterwards.

For a pair, the source orders the effective vectors so that the second is null. With denominator powers j, k , it defines

$$y_{12} = \frac{w v_1 \cdot v_2}{2S_1 S_2}, \quad z_- = \frac{1 - \sqrt{1 - m_1^2}}{2y_{12}}, \quad z_+ = \frac{1 + \sqrt{1 - m_1^2}}{2y_{12}}, \quad (26)$$

$$a_E = j, \quad b_E = 1 - k - \epsilon, \quad c_E = 2 - k - 2\epsilon, \quad (27)$$

$$I_E = \int_0^1 d\tau \tau^{a_E-1} (1 - \tau)^{c_E-a_E-1} [1 - (1 - z_-)\tau]^{-b_E} [1 - (1 - z_+)\tau]^{-b_E}. \quad (28)$$

The prefactors stored with this input are

$$\mathcal{O}_{jk} = 2^{2-j-k-2\epsilon} \pi^{1-\epsilon} \frac{\Gamma(1-k-\epsilon)}{\Gamma(2-k-2\epsilon)}, \quad (29)$$

$$\mathcal{E}_{jk} = \frac{\Gamma(c_E)}{\Gamma(a_E)\Gamma(c_E - a_E)}, \quad (30)$$

$$M_{\text{pair}} = \frac{J_{\text{cut}} \mathcal{O}_{jk} \mathcal{E}_{jk}}{S_1^j S_2^k y_{12}^j} I_E. \quad (31)$$

These are the unexpanded Euler definitions transcribed from the input builder. The map $\tau = v/(1+v)$ puts the integral on the positive half-line used by the tuple. Its Jacobian is included once in the density. If both vectors are null, the source sets $z_- = 0$ in the defining integrand before expansion and sends a distinct class to the engine. A limit of an already expanded massive expression is not used to define that class.

Dimensional orders and algebraic branches. An integral’s finite term is generally insufficient when its Kira coefficient has a pole in ϵ . The input builder measures the Laurent valuation of the reduction coefficients and keeps the required master orders. Endpoint assembly also retains the regulated prefactors and soft powers. The saved root definitions and physical substitutions accompany every evaluated class so that subsequent logarithms and dilogarithms are continued on the same branch.

What is reused across channels. The master value depends on its cut measure and propagator geometry; the diagram, color, spin and charge dependence resides in $c_{c,\Gamma a}$ of Eq. (21). Shared master evaluation is therefore compatible with different channel coefficients. It does not license importing a coefficient or average from a different observed-particle assignment.

3.5 The accepted cut-master inventory

The following entries are transcribed from the shared S08 result. Each tuple lists the four propagator indices in its named family. The first two entries are the physical cuts. A common class means a common Euler input; each master still has its own geometric substitutions and regulated prefactor.

Family	Indices	Euler class
R01	(1, 1, 0, 0)	volume
R01	(1, 1, 1, 1)	pair, both null
R02	(1, 1, 1, 1)	pair, both null
R03	(1, 1, 0, 1)	single, non-null
R03	(1, 1, 1, 1)	pair, one non-null
R04	(1, 1, 1, 1)	pair, one non-null
R09	(1, 1, 1, 1)	pair, one non-null
R10	(1, 1, 1, 1)	pair, one non-null

For the powers occurring in this inventory, the saved half-line densities supplied to **STIntegrate** are particularly short:

$$f_{\text{volume}}(v) = \frac{2^{1-2\epsilon}(1+v)^{-2+2\epsilon}}{v^\epsilon}, \quad (32)$$

$$f_{\text{single}}(v; z_*) = \frac{(1+v)^{-1+2\epsilon}(1+z_*)}{4^\epsilon v^\epsilon (v+z_*)}, \quad (33)$$

$$f_{\text{pair,null}}(v; z_+) = \frac{(1+vz_+)^\epsilon}{1+v}, \quad (34)$$

$$f_{\text{pair,non-null}}(v; z_-, z_+) = \frac{[(1+vz_-)(1+vz_+)]^\epsilon}{1+v}. \quad (35)$$

Each density is integrated with dv on $[0, \infty)$ and multiplied by the corresponding saved prefactor. These are the actual parameter inputs recorded by the tool, not a list of evaluated master values.

4 Soft limits and distributions

A master series at fixed $w > 0$ does not yet define its action at $w = 0$. Shared S15 therefore finds the required soft regions before expanding the regulator. It uses SubTropica's **STGetRegionVectors** and **STPuisseux**, integrates every returned region and keeps the regulated powers of w until distribution assembly.

The saved plus-distribution convention is

$$L_n(w; B) = \left[\frac{\log^n(w/B)}{w} \right]_+, \quad 0 \leq w \leq B, \quad (36)$$

$$\int_0^B dw L_n(w; B) \varphi(w) = \int_0^B dw \frac{\log^n(w/B)}{w} [\varphi(w) - \varphi(0)]. \quad (37)$$

Here B has the same units as w . In the paper's convolution variables its physical upper endpoint is

$$B = Q^2 \left(\frac{1}{\hat{x}} - 1 \right) (1 - z) - \frac{P_{hT}^2}{z}. \quad (38)$$

The symbolic coefficient files retain B explicitly.

Shared S17 collects the soft terms of the form $A_\kappa(\epsilon)w^{-1-\kappa\epsilon}$. The delta coefficient is obtained by evaluating the regulated endpoint integral in the program; the logarithmic coefficients are obtained from the exponential series:

$$E_\kappa(\epsilon; B) = \int_0^B dw w^{-1-\kappa\epsilon}, \quad \exp[-\kappa\epsilon \log(w/B)]. \quad (39)$$

The calculation uses the convergent orientation $\kappa > 0$, $\epsilon < 0$ for this endpoint definition, followed by its Laurent expansion. It checks cancellation of stronger recoil singularities and of higher logarithmic plus terms. The saved result separates **Delta**, **L0**, **L1** and the ordinary **Regular** remainder.

The three channels with virtual corrections retain the physical branches

$$t = \sigma\omega - s, \quad \omega > 0, \quad \sigma = +1, -1. \quad (40)$$

The key 0 in the final files is their common boundary $t = -s$, evaluated by symbolic limits. It must be read from that saved branch; direct substitution into separately singular terms is not the prescribed boundary operation.

4.1 The order of the endpoint operations

The program first forms the regulated master representation, then finds its small- w regions. It retains the region scaling exponent together with each coefficient series. Only after multiplying by the channel's reduction coefficients and combining the required regions does it separate delta, plus and ordinary pieces. A fixed- w Taylor expansion alone would not supply the endpoint action in Eq. (36).

For each regulated soft row the endpoint contribution is defined by $A_\kappa(\epsilon)E_\kappa(\epsilon; B)$, while the logarithmic coefficients come from the recorded expansion of $A_\kappa(\epsilon)B^{-\kappa\epsilon}\exp[-\kappa\epsilon\log(w/B)]$. The source performs both Laurent products at the requested order. The ordinary remainder is the expanded interior expression with its expanded singular soft terms removed. Consequently its addition to the distribution pieces recovers the same regulated real integral when applied to a smooth test function.

The endpoint program checks cancellation of powers stronger than the required $w^{-1-\kappa\epsilon}$ structure, and its saved coefficient of L_2 vanishes at this order. For Hgg, Hqqbar and Hqqprime the complete endpoint coefficients vanish in the final assembly. That result is checked on their own contractions, rather than assumed from the absence of a Born process.

4.2 Region enumeration and physical branches

The accepted source explicitly processes every vector returned by `STGetRegionVectors` and calls `STPuisseux` for each region. It does not rely on the installed convenience wrapper that, in the recorded environment, processed only its first returned region. Raw region outputs and the nonempty regulator series are retained with the accepted result. This distinction concerns which engine outputs enter the calculation; it is not an additional analytic approximation.

The branches $s + t > 0$ and $s + t < 0$ encode the physical continuation of the combined expression. The boundary entry is obtained from the common symbolic limit after the relevant terms are combined. A numerical consumer must use the matching branch and the saved boundary operation, and must not repair a singular component by assigning it zero.

5 Virtual integrals and UV renormalization

5.1 Scalar loop families and fresh master values

For Hqq, Hqg and Hgg, FeynCalc contracts each one-loop amplitude with its Born amplitude and expresses the result in complete scalar loop families. Every mapped integrand is reconstructed before integration. The loop measure inherited from the generated amplitude is

$$\frac{\mu^{2\epsilon}}{(2\pi)^D} \int d^D \ell. \quad (41)$$

Kira canonicalizes and reduces the virtual targets. The accepted run maps 1625 original targets to 252 canonical targets and seven scalar masters. The masters use two freshly derived Euler templates, for bubble and box parents. The evaluated series reach ϵ^1 for the bubble template and ϵ^0 for the box template, as required by their coefficients.

FeynCalc constructs the parameter inputs with `FeynmanIntegralPrefactor -> "Unity"`; the generated amplitude's loop measure remains external. SubTropica evaluates the Euclidean master series. Shared S19 then applies the recorded physical substitutions and Feynman prescription

to their logarithms and dilogarithms. Only after this continuation does it form the Hermitian interference,

$$V_\Gamma = I_\Gamma + I_\Gamma^*, \quad I_\Gamma = \sum_a c_{\Gamma a}^V M_a^V. \quad (42)$$

Here I_Γ includes the stored loop measure and pre-Hermitian interference convention.

5.2 Bubble and box parent integrals

The virtual input builder starts from each master’s active propagators, normalizes the routing signs and constructs the matrix of squared differences between its loop-momentum shifts. It groups masters by the zero pattern of that matrix and derives a generic parent Gram matrix. Every nonzero chord has a symbolic Euclidean parameter $c_{ab} > 0$. The map from that parameter to the actual physical invariant is saved separately for each master.

FeynCalc produces a homogeneous projective parameter representation. The program checks its scaling before fixing the final parameter to one. The two resulting inputs, quoted from shared S12, are

$$I_2(c_{12}) = i\pi^{2-\epsilon}\Gamma(\epsilon) \int_0^\infty dx_1 \frac{(1+x_1)^{-2+2\epsilon}}{(c_{12}x_1)^\epsilon}, \quad (43)$$

$$I_4(c_{14}, c_{23}, c_{34}) = i\pi^{2-\epsilon}\Gamma(2+\epsilon) \int_0^\infty dx_1 dx_2 dx_3 \frac{(1+x_1+x_2+x_3)^{2\epsilon}}{(c_{14}x_1 + c_{34}x_3 + c_{23}x_2x_3)^{2+\epsilon}}. \quad (44)$$

They represent unnormalized Minkowski $d^D\ell$ integrals with the stated FeynCalc prefactor convention. The amplitude’s $\mu^{2\epsilon}/(2\pi)^D$ factor is applied separately. SubTropica evaluates these Euler inputs to the orders requested by their measured coefficients.

The seven physical-master substitutions saved in this run are:

Kira family and indices	Parent	Physical parameter map
V01 (0, 1, 0, 1)	I_2	$c_{12} = -s$
V02 (1, 1, 0, 0)	I_2	$c_{12} = -t$
V03 (0, 1, 1, 0)	I_2	$c_{12} = Q^2$
V03 (1, 1, 0, 0)	I_2	$c_{12} = Q^2 + s + t$
V06 (1, 1, 1, 1)	I_4	$(c_{14}, c_{23}, c_{34}) = (-t, -s, Q^2)$
V07 (1, 1, 1, 1)	I_4	$(c_{14}, c_{23}, c_{34}) = (-s, Q^2 + s + t, Q^2)$
V08 (1, 1, 1, 1)	I_4	$(c_{14}, c_{23}, c_{34}) = (-t, Q^2 + s + t, Q^2)$

These substitutions accompany the recorded Feynman prescription; their signs alone are not a replacement for that prescription. In particular, continuation is performed before adding the conjugate interference. All 252 canonical targets in eleven families are reduced before the seven-master assembly. Triangle and tensor structures appearing in intermediate diagrams are therefore not additional unevaluated outputs.

5.3 Self energies, scaleless terms and the UV/IR separation

Self energies and UV residues are also evaluated with Kira and SubTropica. Shared S14 uses generated quark, gluon/ghost and one-flavor gluon two-point functions. It isolates the logarithmic large-loop-momentum terms with an auxiliary mass, reduces them with Kira, and evaluates the auxiliary vacuum and off-shell bubble masters with SubTropica. The auxiliary mass is an infrared rearrangement device; it is not a physical quark mass.

In the stored convention, with the strong couplings removed from the interference, the accepted common coupling residue is

$$R_g = -\frac{11N_c - 2N_f}{96\pi^2}, \quad \Delta_{\overline{\text{MS}}} = \frac{1}{\epsilon} - \gamma_E + \log(4\pi). \quad (45)$$

This quotes `CouplingResidue` and `MSPole` from the new S14 result. The generated Born coupling count gives `CouplingWeight=2` in each of the three Born channels. The corresponding stored coupling counterterm is

$$V_{\Gamma,c}^{\text{CT}} = n_{g,c} R_g \Delta_{\overline{\text{MS}}} B_{\Gamma,c}^{(D)}, \quad n_{g,c} = 2, \quad (46)$$

with the common-regulator and external-field terms included by S19. The program keeps UV and IR residues separate long enough to check UV cancellation. Kira's scaleless on-shell boundary and the generated external field counts fix the external-field bookkeeping. They are not imported Package-X results. The UV poles cancel before the new renormalized virtual tensors enter the final real-plus-virtual assembly.

The saved field UV residues in the same stripped-coupling convention are

$$R_q^{\text{UV}} = -\frac{(N_c - 1)(N_c + 1)}{32\pi^2 N_c}, \quad (47)$$

$$R_g^{\text{field,UV}} = -\frac{N_f}{24\pi^2} + \frac{5N_c}{48\pi^2}. \quad (48)$$

The gluon result records the gauge-sector contribution and the generated one-flavor fermion contribution separately before multiplying the latter by N_f . These are S14 tool outputs. The coupling residue in Eq. (45) and the gluon field residue in Eq. (48) are different quantities.

For all three Born channels the generated external-field inventory has two quark fields and one gluon field. S19 constructs the external-field factor from the product of the square-root field factors and extracts the interference term at the required order. Its on-shell common-regulator residue obeys the scaleless boundary returned by Kira. UV and IR labels are nevertheless kept separate during the cancellation test, so an on-shell zero does not cause the UV residue needed by renormalization to be silently discarded.

5.4 The renormalized virtual assembly

In the notation of the executed S19 operations, let $V_\Gamma^{\text{bare}} = I_\Gamma + I_\Gamma^*$ and let U_Γ denote the corresponding Hermitian UV residue. The program forms the separated expression

$$V_\Gamma^{\text{sep}} = V_\Gamma^{\text{bare}} + U_\Gamma \left(\frac{1}{\epsilon_{\text{UV}}} - \frac{1}{\epsilon} \right), \quad (49)$$

then adds the generated external-field interference and the coupling counterterm, each multiplying the appropriate D -dimensional Born tensor. It requires the coefficient of $1/\epsilon_{\text{UV}}$ to vanish. Only after this test are the regulator labels identified and the remaining series retained for combination with real emission and factorization. IR poles at this boundary are expected components of that subsequent combination; a virtual tensor is not required to be IR finite by itself.

This ordering also fixes the finite $\overline{\text{MS}}$ constants. The pole factor contains $-\gamma_E + \log(4\pi)$ in the same convention as the loop measure and the collinear subtraction. A standalone textbook UV coefficient with a different stripped-coupling or measure convention cannot be substituted for this complete assembly.

6 Collinear factorization and finite assembly

6.1 The two factorization boundaries

The paper’s factorization prescription removes overlaps with the perturbative parton distributions and fragmentation functions. In its splitting-kernel convention, Eq. (46) can be written as

$$\begin{aligned}\widehat{W}_{ij}^{\text{NLO}} &= W_{ij,\text{unsub}}^{\text{NLO}} + \frac{\alpha_s}{4\pi} \frac{S_\epsilon}{\epsilon} \sum_a \int \frac{d\eta}{\eta} \widehat{W}_{aj}^{\text{LO}}(q, \hat{x}/\eta, \hat{z}) P_{a/i}(\eta) \\ &+ \frac{\alpha_s}{4\pi} \frac{S_\epsilon}{\epsilon} \sum_b \int \frac{d\eta}{\eta^2} \widehat{W}_{ib}^{\text{LO}}(q, \hat{x}, \hat{z}/\eta) P_{j/b}(\eta).\end{aligned}\quad (50)$$

The integration ranges and endpoint actions are those of the paper’s convolutions. S_ϵ/ϵ denotes its $\overline{\text{MS}}$ subtraction pole. The plus signs here follow the negative pole in the perturbative PDF/FF definitions in the paper. The saved **Counterterms** or **PhaseCT** inputs already include their signs.

The Born tensor inside these convolutions is D dimensional. For an incoming rescaling, the $p^\mu p^\nu$ contraction also needs the saved η^{-2} projector conversion. The kernel normalizations, fixed-flavor P_{qg} convention and any quark/antiquark multiplicities are retained from each channel’s own subtraction inputs.

For Hqq, Hqg and Hgq, the loop-independent subtraction results are reused unchanged. For Hgg, Hqqbar and Hqqprime, shared S16 evaluates the preserved Born convolution and projector definitions afresh. Neither operation loads an old integrated real or virtual tensor.

6.2 Kernels, roots and projector rescaling

The core subtraction inputs retain the following kernel convention, quoted here using v for the splitting fraction and $\mathcal{P}(v) = [1/(1-v)]_+$:

$$P_{qq}(v) = - \frac{(N_c - 1)(N_c + 1)}{2N_c} [2 + 2v - 3\delta(1-v) - 4\mathcal{P}(v)], \quad (51)$$

$$P_{qg}(v) = 1 - 2v + 2v^2, \quad (52)$$

$$P_{gq}(v) = \frac{(N_c - 1)(N_c + 1)(2 - 2v + v^2)}{N_c v}, \quad (53)$$

$$\begin{aligned}P_{gg}(v) &= - \frac{1}{3v} [-12N_c + 24N_c v - 12N_c v^2 + 12N_c v^3 \\ &+ 2N_f v \delta(1-v) - 11N_c v \delta(1-v) - 12N_c v \mathcal{P}(v)].\end{aligned}\quad (54)$$

This notation reproduces the stored kernel definitions without redistributing their normalization among the coupling prefactor, species count and convolution. The metadata additionally specifies kernel-normalization weights. In the Hqq and Hgq core inputs these are 1/2 for P_{qq} , P_{gq} , P_{gg} and 1/4 for P_{qg} , together with the recorded fermion-species multiplicity 2. Hqg records 1/2 for its kernel normalization. These factors belong to those specific contraction and fixed-flavor inputs. They must be used together with their stored counterterm definitions, rather than appended to an independently normalized splitting formula.

The Born delta function fixes the splitting fraction in an interior real configuration. The accepted core subtraction output records

$$\eta_{\text{PDF}} = -\frac{t}{w-t}, \quad \eta_{\text{FF}} = \frac{s-w}{s}, \quad (55)$$

$$J_{\text{PDF}} = \frac{1}{w-t}, \quad J_{\text{FF}} = \frac{s-w}{s^2}, \quad S_{pp,\text{PDF}} = \eta^{-2}. \quad (56)$$

The root and Jacobian are obtained from the defining rescaled Born kinematics in the source program. The last factor converts the incoming pp contraction to the original incoming momentum. The metric contraction has no such projector rescaling. These details matter independently of the splitting kernel.

Kernel delta and plus terms are transformed with their endpoint action. The resulting subtraction records carry their own **Delta**, **L0** and ordinary pieces, including finite terms from the D -dimensional Born tensor. They are combined with the real and virtual objects before taking the finite part.

6.3 Restoring the hard-tensor normalization

For the three Born channels, the executed assembly has the form

$$\mathbf{T}_c^{\text{NLO}} = \left[\mu^{2\epsilon} N_{\text{ps}}(D) \mathbf{R}_c^{\text{cut}} + 2\pi \mathbf{V}_c^{\text{ren}} \delta(w) + \mathbf{C}_c^{\text{coll}} \right]_{\epsilon^0}, \quad (57)$$

$$\hat{F}_{i,c}^{\text{NLO}} = e_c^2 g_s^4 N_H \mathcal{P}_i^{(4)} [\mathbf{T}_c^{\text{NLO}}], \quad g_s^2 = 4\pi\alpha_s. \quad (58)$$

The bold quantities have g and pp components; e_c here is the fixed quark charge of the Born process, including Hgq. In Hqq, $\mathbf{R}_c^{\text{cut}}$ includes the saved flavor-weighted sum of real sectors. The stripped-coupling convention in these equations is the one used by the corresponding S20 core-channel code.

The three channels without Born terms retain their couplings in the imported real tensors, so S20 uses their own stored normalization instead of multiplying them by the core-channel prefactor again. Their spectator weight, phase and collinear term are combined before the same finite projector operation. The code symbols **mu** and **mu2** belong to the respective saved scale conventions; **mu2** denotes a squared scale. Comparisons and numerical consumers use the recorded scale maps.

The finite-part notation in Eq. (57) is used only after the program has required

$$[\epsilon^{-n}] (\mathbf{R} + \mathbf{V} + \mathbf{C}^{\text{coll}}) = 0 \quad \text{for every present } n > 0, \quad (59)$$

with all normalization factors included. This check is made separately for the two contractions and their saved distribution and charge components. Taking $\epsilon = 0$ before this combination would lose the finite terms generated by poles multiplying D -dependent factors.

The Born contribution. For the stripped core Born contractions $\mathbf{B}_c^{(D)}$, the executed final projection is

$$\hat{F}_{i,c}^{\text{LO}} = e_c^2 (4\pi\alpha_s) 2\pi N_H \mathcal{P}_i^{(4)} [\mathbf{B}_c^{(D)}|_{D=4}] \delta(w). \quad (60)$$

This quotes the production assembly operation. The explicit Born tensors in the next sections make its channel content unambiguous. The complete NLO coefficient uses Eq. (58), with its coupling and charge factors already restored in the final files.

The acceptance boundary. No negative Laurent coefficient is discarded merely because a finite term can be extracted. UV cancellation is required in the virtual route, then final IR cancellation is required in each contracted, weighted distribution component. Core boundary branches must agree under their saved common-limit operation. Only the accepted finite expressions are exported as the final F hats.

7 Hqq: an observed quark of the incoming flavor

7.1 Processes and the Born normalization

The incoming state is $\gamma^*(q)q_i(p)$ and the tagged parton is $q_i(k_1)$. The two generated Born diagrams form the QCD Compton amplitude with recoil $g(k_2)$. Its virtual correction has the same external state. The core input uses the quark average A_q of Eq. (9) and removes $e_i^2 g_s^2$ from the Born contractions.

The following equations quote `BornPg` and `BornPpp` from this channel's imported `s02_result.wl`. To keep the polynomial readable, its numerator is displayed separately:

$$\mathcal{N}_{qq} = 4Q^4 + 4Q^2s - 2s^2 + Ds^2 + 4Q^2t - 8st + 2Dst - 2t^2 + Dt^2, \quad (61)$$

$$B_{g,qq}^{(D)} = \frac{(D-2)(N_c-1)(N_c+1)\mathcal{N}_{qq}}{2sN_ct}, \quad (62)$$

$$B_{pp,qq}^{(D)} = \frac{(D-2)(N_c-1)(N_c+1)(Q^2+s+t)}{2N_c}. \quad (63)$$

The stored photon and gluon Ward contractions vanish. These Born tensors are used both in the LO projection of Eq. (60) and inside the pole-multiplied NLO terms. Auxiliary Hgq and Hqq Born contractions are generated for the mixed factorization routes.

7.2 Real sectors and their flavor weights

All three real sectors in the process table are required, together with the virtual correction to $\gamma^*q_i \rightarrow q_i g$. The saved generated tag weights are 1/2 for the $q_i gg$ sector and 1 for each of the same-flavor and distinct-flavor quark sectors. These weights belong to the imported tensor convention and must not be applied a second time at final assembly.

The distinct-flavor charge polynomial is retained. Its saved charge moments have the definition

$$\mathcal{Q}_n^{(i)} = \sum_{f=1}^{N_f} e_f^n (1 - \delta_{if}), \quad n = 1, 2. \quad (64)$$

The stored observed-flavor label is $i = 1$, with the accepted integer domain $N_f \geq 1$. The per-sector flavor weights and these moments accompany the final payload. The PDF subtraction uses the Hqq Born tensor with P_{qq} and the Hgq Born tensor with P_{qq} . The FF subtraction uses the Hqq Born tensor with P_{qq} and the Hqg Born tensor with the fixed-flavor P_{qg} .

More explicitly, the real assembly's charge-component dictionary uses

$$\mathbf{R}_{qq}^{\text{cut}} = \mathbf{R}_{qgg,0} + \mathbf{R}_{\text{same},0} + n_{\text{other}} \mathbf{R}_{\text{distinct},0} + \frac{\mathcal{Q}_1^{(i)}}{e_i} \mathbf{R}_{\text{distinct},1} + \frac{\mathcal{Q}_2^{(i)}}{e_i^2} \mathbf{R}_{\text{distinct},2}, \quad (65)$$

where the saved flavor weight is

$$n_{\text{other}} = \begin{cases} N_f - 1, & N_f \geq 2, \\ 0, & \text{otherwise,} \end{cases} \quad N_f \in \mathbb{Z}, \quad N_f \geq 1. \quad (66)$$

The component subscripts are the stored powers of the other flavor's charge after removing the fixed Born charge. The overall e_i^2 in Eq. (58) restores the physical charge convention. The moment definitions, weights and domain are part of the result, so a consumer must not substitute a single value of N_f for all three charge structures.

Each sector goes through the same two physical cuts and the shared Kira masters. Its rational coefficients, interference signs and tag weights remain distinct. The quark-pair sectors include the photon attachments and their interference before charge summation; neither a flavor count nor a guessed gluon-splitting term replaces those generated tensors.

7.3 Subtraction, virtual assembly and final output

The channel's factorization routes are

Boundary	Born tensor	Splitting kernel
Incoming PDF	Hqq	$P_{q/q}$
Incoming PDF	Hgq	$P_{g/q}$
Outgoing FF	Hqq	$P_{q/q}$
Outgoing FF	Hqg	fixed-flavor $P_{q/g}$

The incoming pp route uses the projector conversion of Eq. (56). The virtual contribution uses the channel's own Born interference and the common new self-energy residues with its generated field count. All three real sectors, that virtual term and the four subtraction routes enter the distribution-level pole test.

The final output has an LO delta coefficient and NLO delta, L_0 , L_1 and ordinary coefficients for both $\hat{F}_1$ and $\hat{F}_2$. Its final stage is `Hqq/s17_result/`. The two physical branches and their common boundary retain the charge-moment information of Eq. (65). In hadronic numerics this result is paired with same-flavor quark or antiquark PDFs and FFs, using the saved charge-conjugation contract.

8 Hqg: an observed gluon from an incoming quark

8.1 Observed-leg assignment and Born tensor

The observed gluon is k_1 ; the recoil is $q_i(k_2)g(k_3)$ in the real term and $q_i(k_2)$ in the Born/virtual term. The generated quark average, charge removal and tagged-real weight are retained. This placement of the observed leg controls the fragmentation subtraction. The input kernels are P_{qq} on the PDF side, and P_{gg} and P_{gq} for the gluon and quark parents on the FF side. The quark charge remains fixed; N_f is the closed massless-flavor count.

The photon and incoming quark are the same species as in Hqq, but the observed field is now the gluon. FeynArts generates the Born, real and one-loop amplitudes with this external ordering. The saved Born contractions, with $e_i^2 g_s^2$ removed, are

$$\mathcal{N}_{qq} = -2Q^4 + DQ^4 + 4Q^2s + 4s^2 - 8Q^2t + 2DQ^2t + 4st - 2t^2 + Dt^2, \quad (67)$$

$$B_{g,qq}^{(D)} = -\frac{(D-2)(N_c-1)(N_c+1)\mathcal{N}_{qq}}{2sN_c(Q^2+s+t)}, \quad (68)$$

$$B_{pp,qq}^{(D)} = -\frac{(D-2)(N_c-1)(N_c+1)t}{2N_c}. \quad (69)$$

The source's Born Ward checks pass. These expressions are quoted from the channel's own accepted Born record. They are not obtained in this report by exchanging labels in Eq. (63).

8.2 Real integration and gluon fragmentation

The generated real process is

$$\gamma^*(q)q_i(p) \rightarrow \underline{g}(k_1)q_i(k_2)g(k_3). \quad (70)$$

The cut momenta are k_2 and k_3 . The tagged-square bookkeeping already accounts for observing a gluon among the generated final species. The observed gluon is not integrated as a third cut and no additional spectator symmetry weight is appended to this core tensor. Kira reduces the contracted real coefficients in the common recoil geometry, and the physical cut masters supply their integrals.

The factorization table is

Boundary	Born tensor	Splitting kernel
Incoming PDF	Hqg	$P_{q/q}$
Outgoing FF, gluon parent	Hqg	$P_{g/g}$
Outgoing FF, quark parent	Hqq	$P_{g/q}$

The last row describes a quark produced in the auxiliary Born process whose collinear fragmentation produces the observed gluon. This is why the Hqg subtraction is not the Hqq FF subtraction with an unchanged kernel label.

8.3 Virtual term, finite hats and the consumer

The one-loop correction is interfered with this channel's gluon-tagged Born amplitude. It uses the seven shared virtual masters and the new Kira/SubTropica UV input, with two external quark fields and one gluon field in the recorded count. The renormalized virtual delta term is combined with the real distribution and the three subtraction routes before final projection.

The final `Hqg/s17_result/` contains both LO and NLO hats with the core distribution and branch schema. Its fixed quark charge is supplied by the incoming quark or antiquark in the numerical flavor sum. The final-state FF is $D_{h/g}$; no final quark-flavor sum is hidden in that gluon fragmentation factor.

9 Hgq: an observed fixed-flavor quark from an incoming gluon

9.1 Photon–gluon fusion and the Born tensor

The real recoil is $\bar{q}_i(k_2)g(k_3)$; the Born/virtual recoil is $\bar{q}_i(k_2)$. The incoming average uses the generated D -dimensional physical gluon polarization and color convention. It is not replaced by the incoming-quark average. The PDF routes use the Hgq Born tensor with P_{gg} and the Hqq Born tensor with the per-flavor P_{qg} . The FF route uses the Hgq Born tensor with P_{qq} . The observed quark charge is fixed, while N_f counts closed flavors.

The generated Born process is $\gamma^*g \rightarrow \underline{q}_i\bar{q}_i$. Its saved contractions, with the fixed $e_i^2g_s^2$ factor removed, are

$$\mathcal{N}_{gq} = -2Q^4 + DQ^4 - 8Q^2s + 2DQ^2s - 2s^2 + Ds^2 + 4Q^2t + 4st + 4t^2, \quad (71)$$

$$B_{g,qq}^{(D)} = \frac{\mathcal{N}_{gq}}{t(Q^2 + s + t)}, \quad (72)$$

$$B_{pp,gq}^{(D)} = \frac{4s}{D-2}. \quad (73)$$

The Born normalization includes the generated gluon spin and color average. In particular, the remaining D dependence in $B_{pp,gq}^{(D)}$ is retained inside the factorization and UV products. The Born photon and gluon Ward contractions vanish in the accepted input.

9.2 Real radiation and the subtraction routes

The real process is

$$\gamma^*(q)g(p) \rightarrow \underline{q}_i(k_1)\bar{q}_i(k_2)g(k_3). \quad (74)$$

The observed quark is fixed before the recoil integration. The incoming gluon is summed and averaged according to its own physical polarization projector. The unobserved antiquark and gluon are the two cut lines. No sum over distinct observed flavors is made inside this fixed-flavor hard tensor.

The subtraction inputs retain these routes:

Boundary	Born tensor	Splitting kernel
Incoming PDF, gluon parent process	Hgq	$P_{g/g}$
Incoming PDF, quark Born process	Hqq	per-flavor $P_{q/g}$
Outgoing FF	Hgq	$P_{q/q}$

The per-flavor normalization and fermion-species metadata of the incoming mixing term are used together. The auxiliary Hqq tensor is generated and checked for that contraction convention. Replacing the whole incoming route by a flavor-summed kernel would change the meaning of the fixed observed flavor.

9.3 Virtual correction, output and comparison boundary

The virtual process has recoil $\bar{q}_i$ and is the Born–one-loop interference of photon–gluon fusion. Its loop coefficients differ from Hqq and Hgq even though the scalar master templates are shared. The new UV residues enter with the generated external-field count, and its renormalized term is combined with the cut real result and the three subtraction routes. The final pole and branch tests precede the $\hat{F}_1, \hat{F}_2$ export in `Hgq/s17_result/`.

The numerical consumer uses the gluon PDF and sums the fixed-flavor hard function over active observed quarks and antiquarks with their charges and FFs. The saved BigTMD coefficient comparison passes within its stated scope. The inherited MadGraph real-emission sign comparison is still unresolved; the final IR pole checks do not close that separate comparison. The numerical figures retain this documented input caveat.

10 Hgg: an observed gluon from an incoming gluon

10.1 Generated process and normalization

This contribution is $\gamma^*g \rightarrow gq_f\bar{q}_f$ at the order calculated. The unobserved quark-antiquark pair is integrated once. The saved spectator weight is 1, and the final charge factor is

$$\text{chargeSum} = \sum_{f=1}^{N_f} e_f^2. \quad (75)$$

Quark and antiquark PDF/FF subtraction routes are kept with their own conventions. Their multiplicities do not introduce a second copy of the integrated spectator pair. Real integration and factorization provide the finite result; there is no virtual term at this order.

The generated real amplitude is squared with an incoming-gluon average and with the gluon k_1 treated as observed. The FeynCalc source also generates the auxiliary quark–gluon Born processes required by the collinear terms. The real scalar record retains its strong coupling and charge factors after removing the common electromagnetic coupling used by the lepton cross section. It explicitly records that the spectator weight has not yet been applied.

The field list is $\{g(k_1), q_f(k_2), \bar{q}_f(k_3)\}$. The unobserved quark and antiquark are different species, giving the saved $S_{\text{spectator}} = 1$. Their two positive-energy constraints form the cut master. The same shared eight-master basis supplies the integral, with this channel’s gluon-averaged real coefficients.

10.2 Flavor sum and collinear mixing

The photon couples to the generated quark line. The fixed representative charge in that tensor is mapped at final assembly to $\sum_f e_f^2$. The incoming gluon PDF and outgoing gluon FF do not perform this electromagnetic sum themselves.

The incoming collinear routes use a quark or antiquark from a gluon splitting, followed by a Born process with an observed gluon. The outgoing routes use photon–gluon fusion followed by fragmentation of the quark or antiquark into the observed gluon. Thus both quark and antiquark routes are present in the subtraction bookkeeping. Their species multiplicity is associated with the factorization routes, not with an extra integration of the same real spectator pair.

10.3 Finite projection and the real-only normalization

For this channel and the two following channels, the final code uses the stored real-only assembly operation

$$\mathbf{T}_c^{\text{fin}} = \left[S_{\text{spectator},c}(\mu^2)^{2\epsilon} N_H N_{\text{ps}}(D) \mathbf{R}_c^{\text{cut}} + \mathbf{C}_c^{\text{phase}} \right]_{\epsilon^0}. \quad (76)$$

Here μ^2 denotes the code’s squared scale `mu2`, and $\mathbf{C}_c^{\text{phase}}$ is the complete stored phase-normalized Born convolution. The real tensor in this equation retains its own couplings. The program maps g_s^2 to $4\pi\alpha_s$, performs the channel’s charge substitution, and applies $\mathcal{P}_i^{(4)}$. The core prefactor of Eq. (58) is not applied again.

The executed Hgg result has zero delta and plus coefficients after assembly and a finite ordinary NLO coefficient. Both hats are in `Hgg/s11_result/`. In the neutral-pion convolution the associated luminosity is the gluon PDF times the gluon FF, multiplied by the charge sum according to the coefficient contract.

11 Hqqbar: an observed antiquark of the incoming flavor

11.1 Identical spectator quarks

The process is $\gamma^* q_i \rightarrow \bar{q}_i(k_1) q_i(k_2) q_i(k_3)$. The incoming flavor and its squared charge, `eq2`, are fixed. The two unobserved quarks are identical. S20 reads the generated field list and computes

$$S_{\text{spectator}} = \frac{1}{\prod_a n_a!}, \quad (77)$$

where n_a counts identical unobserved fields. Its saved value for this channel is 1/2, applied once because the imported real tensor excludes that spectator weight. The initial subtraction uses $q_i \rightarrow g$ followed by $\gamma^* g \rightarrow \bar{q}_i q_i$; the final subtraction uses $\gamma^* q_i \rightarrow g q_i$ followed by $g \rightarrow \bar{q}_i$. There is no extra flavor sum and no virtual contribution.

Both assignments of the identical quark lines are included in the generated amplitude before squaring. Their exchange interference and fermion signs belong to the FeynArts/FeynCalc result. The spectator factor divides the phase-space counting once; it does not remove an amplitude or its interference. The source uses a quark incoming average, while the observed antiquark is summed as the tagged final state.

11.2 Cut integration and subtraction

The real tensor is mapped with $k_2 = r$, $k_3 = P_X - r$ and the same common invariants as the other channels. The two unobserved quarks are the cut lines. The rational master coefficients retain the identical-flavor interference. Kira reduction and SubTropica evaluation then proceed with the shared physical master inputs.

The incoming PDF route changes the incoming quark into a gluon and uses the Born photon–gluon-fusion tensor with the antiquark observed. The FF route uses the Born observed-gluon tensor followed by the fixed-flavor $g \rightarrow \bar{q}_i$ fragmentation kernel. Shared S16 evaluates these preserved Born convolution definitions, including their D -dimensional projectors and flavor normalization. There is no virtual amplitude or UV loop term to add at this order in this channel.

11.3 Charge convention and final hats

The spectator factor 1/2 and the phase normalization enter Eq. (76). The final charge substitution is $\text{eq}^2 \rightarrow \text{eq2}$; eq2 denotes the fixed squared incoming-flavor charge, not a sum over active flavors. The endpoint tests give zero delta, L_0 and L_1 , and the ordinary finite coefficients are exported to `Hqqbar/s11_result/`.

The hadronic luminosity pairs an incoming quark with the FF of its antiquark, and includes the conjugate incoming-antiquark process under the saved consumer convention. Even for a neutral hadron with equal quark and antiquark FFs, this channel remains a separate hard coefficient and is counted once in the six-channel sum.

12 Hqqprime: an observed quark of a different flavor

12.1 The two independent photon charges

The process is $\gamma^* q_i \rightarrow q_j(k_1) q_i(k_2) \bar{q}_j(k_3)$ with $j \neq i$. The saved spectator weight is 1. Both charges are retained independently, with the polynomial structure

$$\hat{F}_{a,qq'} = e_i^2 A_a + e_i e_j B_a + e_j^2 C_a, \quad a = 1, 2. \quad (78)$$

This notation labels the saved charge components; no component is dropped as an integration assumption. The initial subtraction carries e_j^2 through the $\gamma^* g \rightarrow q_j \bar{q}_j$ Born process. The final subtraction carries e_i^2 through $\gamma^* q_i \rightarrow g q_i$, followed by $g \rightarrow q_j$. The result is for a fixed distinct observed flavor; physical flavor sums are performed by the consumer.

The generated amplitude retains photon attachment to the incoming-flavor line and to the distinct-flavor line. Their squares give the two squared charge structures, and their interference gives the mixed charge structure. The labels A_a, B_a, C_a refer to those complete coefficient

components after integration and subtraction, rather than to separately guessed splitting approximations.

12.2 Real master coefficients and factorization

The observed q_j is held fixed and the recoil pair $q_i, \bar{q}_j$ is integrated through the two positive-energy cuts. The distinct flavor condition makes the spectator species different, as recorded by the generated field list. The program retains all three charge coefficients while mapping the real contractions to Kira and assembling their master values. A mixed-charge coefficient is not dropped because a later observable may sum it to zero.

There are two factorization routes with different electromagnetic charges. The PDF route has an incoming splitting $q_i \rightarrow g$ followed by $\gamma^* g \rightarrow q_j \bar{q}_j$, so its Born input carries e_j^2 . The FF route has $\gamma^* q_i \rightarrow g q_i$ followed by $g \rightarrow q_j$, so its Born input carries e_i^2 . The same incoming and outgoing convolution measures of Eq. (50) apply with these fixed-flavor assignments. There is no Born hard part for the channel itself and hence no NLO virtual interference to add.

12.3 Final coefficient and flavor summation

The final assembly uses Eq. (76) with spectator weight one. After the charge-resolved pole and endpoint checks, the output in `Hqqprime/s11_result/` is an ordinary finite NLO coefficient polynomial in the two charges. It is for one fixed ordered pair of different flavors. Summing over observed species belongs to the PDF/FF convolution, with an explicit distinct-flavor condition.

For the neutral-pion numerical setup, the consumer verifies equality of conjugate FFs and the corresponding cancellation of the mixed-charge luminosity. That is a checked property of this numerical flavor sum. It neither changes the saved partonic polynomial nor asserts that its mixed coefficient vanishes for arbitrary fragmentation functions.

13 Final F hats and the files that contain them

For Hqq, Hqg and Hgq the output format is

$$\hat{F}_{i,c}^{\text{LO}} = B_{i,c} \delta(w), \quad (79)$$

$$\hat{F}_{i,c}^{\text{NLO}} = D_{i,c} \delta(w) + C_{i,c}^{(0)} L_0(w; B) + C_{i,c}^{(1)} L_1(w; B) + R_{i,c}(w). \quad (80)$$

The file keys are respectively `L0Delta`, `Delta`, `L0`, `L1` and `Regular`. The NLO coefficients have branch keys 1, -1 and 0, as described above. Their coupling, charge and hard-tensor normalization factors are already included.

For Hgg, Hqqbar and Hqqprime, the executed endpoint checks give zero `Delta`, `L0` and `L1`. Their F-hat files contain the finite NLO ordinary coefficient directly. This is an outcome of the new calculation and its endpoint checks, not a condition imposed on the unintegrated real tensors.

Channel	Full final result relative to this report
Hqq	Hqq/s17_result/s17_result.wl
Hqg	Hqg/s17_result/s17_result.wl
Hgq	Hgq/s17_result/s17_result.wl
Hgg	Hgg/s11_result/s11_result.wl
Hqqbar	Hqqbar/s11_result/s11_result.wl
Hqqprime	Hqqprime/s11_result/s11_result.wl

Each listed directory also contains the corresponding `sNN_F1_hat.wl` and `sNN_F2_hat.wl` files. The full result must accompany those direct expressions because it records the support, distributions, branches, scale convention and input identities. The long analytic coefficient expressions are retained in these symbolic files; the present report explains their construction and interpretation.

14 Stage map, execution and checks

Channel-level stages are consecutive links to shared implementations. The numbers in the following table are *channel* step numbers; the shared source filenames retain their original execution numbers.

Operation	Hqq/Hqg/Hgq	Hgg/Hqqbar/Hqqprime
Import unchanged inputs; define kinematics	s01–s02	s01–s02
Real family mapping and Kira reduction	s03–s04	s03–s04
Physical cut-master inputs; SubTropica evaluation	s06, s08	s05–s06
Real master coefficients; regulated soft regions	s09, s13	s07–s08
Virtual families, reduction, inputs and evaluation	s05, s07, s10–s11	–
New self-energy/UV evaluation	s12	–
Collinear inputs	preserved in s01	s09
Real distribution assembly	s14	s10
Virtual coefficients and renormalized assembly	s15–s16	–
Final pole cancellation and F-hat projection	s17	s11

The table groups operations by purpose; the numbered channel sequence and each README give the actual dependencies. Shared integrations and reductions run once for the applicable channel group.

The accepted calculation ran on Hoffman2 compute nodes. The local folders retain the main sources, results, logs, Kira databases, SubTropica outputs and resumable caches. Independent work is divided among a bounded number of workers with per-operation memory limits, process-tree RSS guards and persistent job monitors. The recorded accepted runs did not encounter OOM.

The saved gates cover input reconstruction, Born normalization and Ward identities, complete Kira target coverage, retention of both cuts, parameter-measure normalization, regulator depth, physical continuation, UV cancellation, final IR cancellation and the required branch boundaries. The all-six final acceptance and its execution receipt are in `common/s20_result/`.

The subsequent `bigTMD_comparison/` compares the frozen new F hats with the same reference used by the earlier comparison workflow. Its saved report gives **60/60 exact comparisons and 120/120 high-precision checks passing**. The exact comparison uses the documented rational reconstruction of printed reference decimals, SU(3), matched charges, scales and distribution conventions. It is a comparison after production, and does not set or tune a coefficient. The scope of the public numerical driver differs from that of the complete reference coefficient files and is recorded separately.

The inherited **Hgq MadGraph real-emission sign comparison remains unresolved**. Neither the passed pole checks nor the BigTMD coefficient comparison is reported as closing that separate check.

15 Numerics: from partonic hats to a hadronic cross section

15.1 Inputs and the quantity being predicted

The separate `numerics/` folder consumes the six new symbolic results [5]. It predicts the neutral-pion transverse-momentum distribution with the saved H1 bin definitions and cuts. The symbolic hard scattering ends at the finite F hats; PDFs, FFs, active-flavor choices and experimental cuts enter here. The accepted numerical run evaluates the new F hats, rather than loading old channel-integrated bin totals.

The paper’s convolution definitions are

$$F_1 = \sum_{ij} \int_x^1 \frac{d\xi}{\xi} \int_z^1 \frac{d\zeta}{\zeta^2} \widehat{F}_{1,ij}(x/\xi, z/\zeta, Q^2, q_T) f_{i/H}(\xi) D_{h/j}(\zeta), \quad (81)$$

$$F_2 = \sum_{ij} \int_x^1 d\xi \int_z^1 \frac{d\zeta}{\zeta^2} \widehat{F}_{2,ij}(x/\xi, z/\zeta, Q^2, q_T) f_{i/H}(\xi) D_{h/j}(\zeta). \quad (82)$$

The different ξ measures are part of the paper’s F_1, F_2 definitions. The fragmentation Jacobian is applied once, and the distributions act on the complete convolution test function.

The azimuth-independent lepton cross section used by the numerical definitions is the paper’s expression

$$\frac{d\sigma}{dx dy dz dp_T^2} = \frac{\pi^2 \alpha_{\text{EM}}^2}{zxyQ^2} [xy^2 F_1 + (1-y)F_2]. \quad (83)$$

The maps and weights needed to express it in the integrated variables are derived in numerical s04 from its defining physics inputs. This report quotes those saved outputs. It does not introduce a different lepton normalization or a different F_2 convolution measure.

15.2 Beam energies, cuts and scale choices

Input	Accepted numerical setting
Electron and proton energies	$E_e = 27.6 \text{ GeV}, E_p = 820 \text{ GeV}$
Inelasticity	$0.1 < y < 0.6$
Photon virtuality panels	$2 < Q^2 < 4.5, 4.5 < Q^2 < 15, 15 < Q^2 < 70 \text{ GeV}^2$
Transverse momentum	H1 bins between 2.5 and 15 GeV
Pion laboratory angle	$5^\circ < \theta_h < 25^\circ$
Pion laboratory energy	$E_h/E_p > 0.01$
Common scale	$\mu_R^2 = \mu_F^2 = \mu_D^2 = (Q^2 + p_T^2)/2$
PDF	MRST2002 NLO, original interpolation, mode 1
FF options	KKP neutral pion; Kretzer neutral-pion average
Hard-part color	$N_c = 3$ in the consumer
Mass and azimuth treatment	Massless kinematics; uniform-azimuth F_1, F_2 approximation

Here μ_D denotes the fragmentation factorization scale. The reported plots use the common central-scale choice above. They do not show an independent scale-variation band. Laboratory cuts are applied to the pion momentum reconstructed from the saved massless frame map, not to the partonic recoil or directly to a centre-of-mass polar angle.

15.3 PDF, FF and coupling conventions

The native MRST interpolation returns the momentum-weighted PDF $\xi f_a(\xi, \mu^2)$. The numerical adapter converts it to the f_a required by the convolution before applying the explicit $1/\xi$ factor for F_1 . This separates the PDF library's output convention from the structure-function measure.

The KKP input uses its saved pion option `ih=5` and NLO `iset=1`. The Kretzer input uses its NLO set and the recorded neutral combination $(\pi^+ + \pi^-)/2$. The adapter checks the neutral quark/antiquark equalities needed by the flavor assembly. It does not identify quark and antiquark PDFs; those remain distinct.

The retained native flavor order is

$$\{g, u, \bar{u}, d, \bar{d}, s, \bar{s}, c, \bar{c}, b, \bar{b}\}. \quad (84)$$

The model supplies the up-type and down-type charges and their antiparticle signs. The numerical hard parts use four active flavors below the saved bottom threshold and five above it; the phase-space contract requires the scale to lie above the charm threshold. The accepted coupling input uses the EPHOX implicit two-loop convention with continuity at the bottom threshold and records

$$\alpha_s(M_Z) = 0.1197, \quad M_Z = 91.188 \text{ GeV}, \quad (85)$$

$$m_c^2 = 2.045 \text{ GeV}^2, \quad m_b^2 = 18.5 \text{ GeV}^2, \quad \Lambda_4^2 = 0.1085488995978293 \text{ GeV}^2. \quad (86)$$

The electromagnetic input is $\alpha_{\text{EM}} = 0.0072973525643$ and the retained unit conversion is $1 \text{ GeV}^{-2} = 389379372.17185944 \text{ pb}$. These are the numerical input records, not additional fitted parameters of the new F-hat calculation.

The PDF and FF adapters are checked on their declared input grids. The retained PDF domain is $10^{-5} \leq \xi \leq 1$ with $1.25 \leq \mu^2 \leq 10^7 \text{ GeV}^2$, and the FF domain is $0.01 \leq \zeta \leq 1$ with $1 \leq \mu^2 \leq 10^6 \text{ GeV}^2$. Evaluation errors are handled as errors by the numerical driver; an unsuccessful library call is not converted to a zero cross-section contribution.

16 Numerics: recoil variables and laboratory acceptance

16.1 From the fragmentation fraction to recoil mass

The integration uses Q^2, y, p_T, z, ξ, w with

$$x = \frac{Q^2}{4E_e E_p y}, \quad \hat{x} = \frac{x}{\xi}, \quad s = -\frac{Q^2(x - \xi)}{x}. \quad (87)$$

The fragmentation fraction is changed to the recoil invariant w . The defining solution recorded in s04, also used in the paper's convolution map, is

$$\zeta(w) = \frac{\hat{x} p_T^2 + z^2 Q^2 (1 - \hat{x})}{z [Q^2 (1 - \hat{x}) - w \hat{x}]} \quad (88)$$

The endpoint value ζ_0 is this map at $w = 0$. The FF must be evaluated separately at $\zeta(w)$ for interior terms and at ζ_0 for delta terms and endpoint subtraction.

The saved lower bound and recoil endpoint are

$$\xi_{\min} = \frac{x(Q^2 z^2 - Q^2 z - p_T^2)}{Q^2 z(z - 1)}, \quad (89)$$

$$B = Q^2 \left(\frac{1}{\hat{x}} - 1 \right) (1 - z) - \frac{p_T^2}{z}, \quad 0 \leq w \leq B. \quad (90)$$

The source solves the remaining support conditions to give the two z bounds

$$z_{\text{low,up}} = \frac{\sqrt{Q^2}(x-1) \pm \sqrt{(x-1)(Q^2x - Q^2 + 4p_T^2x)}}{2\sqrt{Q^2}(x-1)}, \quad (91)$$

where the saved lower root uses the plus sign and the upper root the minus sign. Configurations with empty physical support do not enter the convolution. The resulting maps are checked against the invariant definitions, rather than treating independently sampled s, t, w as an automatically physical point.

16.2 Jacobian and invariant evaluation

The s04 output gives the fragmentation Jacobian explicitly as

$$J_\zeta(w) = -\frac{x[Q^2xz^2 - Q^2\xi z^2 - p_T^2x]}{z[Q^2x - Q^2\xi + wx]^2}. \quad (92)$$

It is the positive integration Jacobian on the accepted support. All new coefficient bundles record that this Jacobian is not already included. The consumer consequently multiplies by J_ζ once, along with the separate $1/\zeta^2$ FF measure.

For completeness, the saved invariant maps passed to the hard functions include

$$t(w) = \frac{Q^2p_T^2\xi + Q^2wxz^2 - p_T^2wx}{Q^2xz^2 - Q^2\xi z^2 - p_T^2x}, \quad (93)$$

$$u(w) = -\frac{Q^2\xi z^2(Q^2x - Q^2\xi + wx)}{x(Q^2xz^2 - Q^2\xi z^2 - p_T^2x)}, \quad (94)$$

$$\hat{z}(w) = \frac{z^2(Q^2x - Q^2\xi + wx)}{Q^2xz^2 - Q^2\xi z^2 - p_T^2x}. \quad (95)$$

These formulas are quoted from the symbolic map result. The branch selector evaluates $s + t$ at the same point, and the saved boundary entry is used for its common limit.

The s04 weights multiplying the structure functions after changing to the integration variables are

$$W_1 = \frac{\pi^2\alpha_{\text{EM}}^2 p_T}{2E_e E_p Q^2 z}, \quad (96)$$

$$W_2 = -\frac{\pi^2\alpha_{\text{EM}}^2 p_T(4E_e E_p y - 4E_e E_p)}{2E_e E_p (Q^2)^2 y z}. \quad (97)$$

The native paired evaluator uses W_1/ξ with $\hat{F}_1$ and W_2 with $\hat{F}_2$. Their different incoming measures are therefore preserved explicitly.

16.3 The pion momentum in the laboratory

The laboratory acceptance is obtained from the saved massless frame transformation. With ϕ the azimuth used in that map, the output is

$$E_h = \frac{2\sqrt{Q^2}p_T z \sqrt{1-y} \cos \phi + Q^2 z^2 + p_T^2 + y z^2 (4E_e^2 y - Q^2)}{4E_e y z}, \quad (98)$$

$$p_{h,x} = \sqrt{Q^2} z \sqrt{1-y} + p_T \cos \phi, \quad (99)$$

$$p_{h,y} = p_T \sin \phi, \quad (100)$$

$$p_{h,z} = \frac{2\sqrt{Q^2}p_T z \sqrt{1-y} \cos \phi + Q^2 z^2 + p_T^2 - y z^2 (4E_e^2 y + Q^2)}{4E_e y z}. \quad (101)$$

The cuts are imposed on E_h and on the massless polar-angle definition $\cos \theta_h = p_{h,z}/E_h$. The source solves the energy and angle inequalities as bounds on $\cos \phi$, clips those bounds to $[-1, 1]$ and calculates the allowed azimuth fraction. An empty interval gives zero acceptance.

The defining quantity is

$$\mathcal{A}_{\text{lab}}(Q^2, y, z, p_T) = \frac{1}{2\pi} \int_0^{2\pi} d\phi \mathbf{1}_{E_h/E_p > 0.01} \mathbf{1}_{5^\circ < \theta_h < 25^\circ}. \quad (102)$$

The accepted numerical source uses the analytic angular fraction derived from these inequalities, so it does not need a seventh Monte Carlo variable for ϕ . It assumes the retained F_1, F_2 contribution is uniform in this azimuth. Azimuth-dependent SIDIS structure functions are outside the plotted approximation; the acceptance fraction alone does not supply them.

17 Numerics: assembling all six channel luminosities

Write $f_a = f_{a/H}(\xi, \mu^2)$ and $D_a = D_{h/a}(\zeta, \mu^2)$. Let a run over active quarks and antiquarks, and let e_a carry the charge sign from the model. The native consumer factors the charge polynomials according to each export contract and constructs the following luminosities. The formulas are the definitions used by numerical s08/s09; when the full hat already contains its charge, the exporter exposes the corresponding normalized charge component so the factor is not included twice.

17.1 Hgg, Hgq, Hqg and Hqqbar

$$\mathcal{L}_{gg} = f_g D_g \sum_{f \text{ quark flavors}} e_f^2, \quad (103)$$

$$\mathcal{L}_{gq} = f_g \sum_a e_a^2 D_a, \quad (104)$$

$$\mathcal{L}_{qg} = D_g \sum_a e_a^2 f_a, \quad (105)$$

$$\mathcal{L}_{q\bar{q}} = \sum_a e_a^2 f_a D_{\bar{a}}. \quad (106)$$

The first sum is over quark flavors, not a doubled quark-plus-antiquark copy of the Hgg spectator pair. In the other sums quarks and antiquarks are distinct incoming or observed species. The neutral-FF equalities are checked where used; they do not erase those species definitions.

17.2 Hqq charge moments

Hqq uses same-species luminosities $f_a D_a$ and evaluates its coefficient with the fixed charge and the other-flavor moments appropriate to that species. The accepted neutral consumer groups equal-charge species using its verified charge-conjugation identities. For each group,

$$\mathcal{L}_{qq, \text{group}} = \sum_{a \in \text{group}} f_a D_a. \quad (107)$$

The hard coefficient is evaluated with its stored $e_i, Q_1^{(i)}, Q_2^{(i)}$ arguments. The saved tool-generated values used by the two active-flavor settings are:

N_f	Representative charge	$\mathcal{Q}_1^{(i)}$	$\mathcal{Q}_2^{(i)}$
4	2/3	0	2/3
4	-1/3	1	1
5	2/3	-1/3	7/9
5	-1/3	2/3	10/9

These values belong to the numerical flavor model. Production Hqq continues to retain symbolic moments and its stated N_f domain.

17.3 Distinct-flavor charge components

For Hqqprime, the pair (a, b) is ordered and its quark-flavor names must differ, independently of whether a species is a quark or an antiquark. The source constructs

$$\mathcal{L}_{\text{incoming}} = \sum_{\text{flavor}(a) \neq \text{flavor}(b)} e_a^2 f_a D_b, \quad (108)$$

$$\mathcal{L}_{\text{prime}} = \sum_{\text{flavor}(a) \neq \text{flavor}(b)} e_b^2 f_a D_b, \quad (109)$$

$$\mathcal{L}_{\text{mixed}} = \sum_{\text{flavor}(a) \neq \text{flavor}(b)} e_a e_b f_a D_b. \quad (110)$$

They multiply the corresponding three charge coefficients of Section 12. The neutral-pion input passes the symbolic and native mixed-luminosity cancellation checks. The mixed partonic coefficient remains in the symbolic result. No factor based only on the number of active flavors replaces these ordered sums.

18 Numerics: distributions, precision and bin integration

18.1 The complete endpoint test function

For a distribution component d and a channel/charge component c , the paired hard evaluator forms the saved combination

$$K_{c,d}(w) = \frac{J_\zeta(w)}{\zeta(w)^2} \left[\frac{W_1}{\xi} \widehat{F}_{1,c,d}(w) + W_2 \widehat{F}_{2,c,d}(w) \right]. \quad (111)$$

The channel's compatible charge luminosity is $\mathcal{L}_c(w)$, using the FF at $\zeta(w)$. In this notation $G_{c,d}(w) = K_{c,d}(w) \mathcal{L}_c(w)$ names the complete test-function product used by the native source. The product includes the kinematic dependence of the coefficient, its Jacobian, the FF and the PDF, rather than subtracting only the value of the analytic F-hat expression.

For a core channel the implemented recoil integration is represented by

$$\begin{aligned} \mathcal{I}_c^{\text{LO}} &= G_{c,\text{Born}}(0), \\ \mathcal{I}_c^{\text{NLO}} &= G_{c,\Delta}(0) + \int_0^B dw \left\{ G_{c,\text{Regular}}(w) + \frac{G_{c,L_0}(w) - G_{c,L_0}(0)}{w} \right. \\ &\quad \left. + \frac{G_{c,L_1}(w) - G_{c,L_1}(0)}{w} \log \frac{w}{B} \right\}. \end{aligned} \quad (112) \quad (113)$$

This is the numerical source's implementation of the saved distribution convention. Its delta terms are evaluated at the endpoint without a recoil integration Jacobian. The ordinary and endpoint-subtracted interior terms carry that Jacobian. Hgg, Hqqbar and Hqqprime enter with their ordinary NLO terms and no LO contribution.

18.2 Six-dimensional sampling map

Each accepted bin uses six unit-cube variables. The transformations are logarithmic for Q^2, p_T, z, ξ and linear for y, w , with the kinematic lower and upper limits evaluated in their dependency order. The definitions in the source are

$$x_{\text{lin}}(v) = a + v(b - a), \quad J_{\text{lin}} = b - a, \quad (114)$$

$$x_{\text{log}}(v) = a \exp[v \log(b/a)], \quad J_{\text{log}} = x_{\text{log}}(v) \log(b/a), \quad 0 \leq v \leq 1. \quad (115)$$

At a sampled point the outer weight contains the five non-recoil Jacobians, $\mathcal{A}_{\text{lab}}$ and the pb conversion. The linear w Jacobian multiplies only the interior terms in Eq. (113). The endpoint contribution is independent of the sixth coordinate and is included with its unit-cube measure.

The PDF is evaluated once for the sampled ξ, μ^2 and used in all channel components. FFs are evaluated at both the interior and endpoint fragmentation fractions. All six channels and both FF choices use the same sample points, retaining the covariance of their sum. Empty physical support or failed experimental cuts produce a defined zero contribution. A failed hard-function evaluation is an error and is not treated as an empty phase-space point.

18.3 Stable evaluation of symbolic coefficients

The symbolic export retains all distribution, branch, coupling, scale and charge contracts. The numerical build contains 110 scalar coefficients arranged into 55 F_1, F_2 pairs. The native evaluator uses the recorded multiprecision complex arithmetic, with the 100-decimal-digit Boost implementation and a high-precision dilogarithm path. Opposite large terms are combined at the prescribed precision before returning the real observable contribution.

Two saved comparison stages validate that translation. Numerical s07 records 7662 native/Wolfram checks. Numerical s10 records 1274 independent checks using 100- and 140-digit evaluation. These are coefficient-evaluation tests against the new symbolic inputs; they are separate from the statistical convergence of an integrated bin.

18.4 VEGAS iterations and statistical acceptance

The run uses VEGAS 6.3 with a recorded seed for every bin. Its initial configuration has three adaptation iterations, then eight production iterations with 1000 requested evaluations per iteration and adaptation disabled. If the acceptance conditions fail, the driver increases the requested evaluations by the recorded doubling rule, up to its bounded refinement limit. Accepted outputs preserve the actual evaluation counts and iteration records.

The final estimator uses the arithmetic average of production iterations, as recorded by `weighted=False`. The joint result retains all 24 signed components: two FF options, six channels and LO/NLO orders. Their covariance is used for the uncertainty of the total; signed channel contributions are not converted into positive weights. The weighted iteration estimate is used only for its consistency diagnostic, not silently substituted for the reported estimator.

The recorded error requirement for each FF prediction is

$$\delta\sigma \leq \max\left(0.05|\sigma|, 0.005 \left. \frac{d\sigma}{dp_T} \right|_{\text{data}} \Delta p_T\right). \quad (116)$$

Additional gates require finite covariance, consistent nonnegative variances, agreement of the first and second iteration halves within the saved three-error pull, a chi-square tail probability at least 0.001, and reconstruction of each FF total from its channel components. All 17 accepted

bins pass their saved gates. Those gates establish numerical integration quality for the specified setup; they are not a test of agreement with the experimental central value.

The plotted quantity is the bin integral divided by its transverse momentum width,

$$\left\langle \frac{d\sigma}{dp_T} \right\rangle_b = \frac{\sigma_b}{\Delta p_{T,b}}. \quad (117)$$

The integration error is divided by the same width. The calculation integrates over the full Q^2, y, p_T bin and over the convolution variables with acceptance, rather than evaluating the hard function at one representative point.

19 Numerical results, plots and comparison conventions

19.1 The saved bin averages

Table 1 transcribes the saved SIDIS plot result. H1 uncertainties are the recorded total experimental errors. KKP and Kretzer uncertainties are numerical integration errors only. Values are rounded for display; the JSON results retain the accepted precision. All cross-section entries in the table have units pb/GeV.

Table 1: Accepted SIDIS bin averages and saved H1 data.

Q^2 [GeV ²]	p_T [GeV]	H1	SIDIS KKP	SIDIS Kretzer
2–4.5	2.5–2.8	215 ± 36	199.9 ± 3.47	142.6 ± 2.71
	2.8–3.3	146 ± 27	130.1 ± 3.19	95.75 ± 3.08
	3.3–4	60 ± 14	64.9 ± 1.32	47.29 ± 1.03
	4–5.2	21.7 ± 4.7	24.31 ± 0.34	16.94 ± 0.254
	5.2–8	4.37 ± 0.83	5.264 ± 0.0767	3.665 ± 0.0526
	8–15	0.29 ± 0.26	0.4058 ± 0.00633	0.2894 ± 0.00465
4.5–15	2.5–2.8	200 ± 34	187.7 ± 3.18	139.3 ± 2.55
	2.8–3.4	122 ± 21	120.9 ± 1.81	88.54 ± 1.42
	3.4–4.1	58 ± 11	64.04 ± 1.06	45.91 ± 0.801
	4.1–5.2	23 ± 4.9	26.31 ± 0.441	18.7 ± 0.338
	5.2–8	5.8 ± 1.2	6.189 ± 0.108	4.409 ± 0.0859
	8–15	0.31 ± 0.12	0.5181 ± 0.00849	0.3713 ± 0.00608
15–70	2.5–2.9	118 ± 18	100.3 ± 2.6	77.83 ± 2.41
	2.9–3.5	69.1 ± 8.8	67.78 ± 1.32	52.85 ± 1.23
	3.5–4.7	27.7 ± 4.7	30.69 ± 0.424	23.65 ± 0.339
	4.7–8	5.41 ± 0.96	6.817 ± 0.122	5.125 ± 0.0962
	8–15	0.21 ± 0.06	0.4823 ± 0.00618	0.3536 ± 0.00454

The corresponding accepted integrated cross sections are given in Table 2. The final row sums the measured bin domains included in these three panels; it is not an extrapolation outside their cuts and transverse-momentum range.

19.2 What the plotted comparisons mean

The solid blue and orange predictions in Figure 1 use the new SIDIS hats with KKP and Kretzer fragmentation, respectively. They are LO plus NLO bin averages with all six channels. The black points are the saved H1 measurements with total experimental errors. The complete BigTMD

Table 2: Integrated cross sections in pb, with integration errors.

Q^2 [GeV ²]	SIDIS KKP	SIDIS Kretzer
2–4.5	217.2 ± 2.17	156.4 ± 1.92
4.5–15	223.6 ± 1.72	162.6 ± 1.35
15–70	143.5 ± 1.46	110.6 ± 1.32
All included bins	584.3 ± 3.13	429.6 ± 2.69

predictions use the recorded full coefficient formulas and the matched numerical inputs. Only the full six-channel BigTMD variant is present in these SIDIS figures.

The dashed paper curves are the preserved original EPS curve vertices identified by the plotting inputs: hep-ph/0411212, Fig. 4, and arXiv:1808.04396, Fig. 3. They are displayed as curves. Their values used in the ratio figure are interpolated at each bin centre, whereas the SIDIS and BigTMD predictions are bin averages. A curve-to-bin-centre ratio is therefore not a reconstruction of the paper’s bin-integrated prediction, particularly for the broadest transverse-momentum bins.

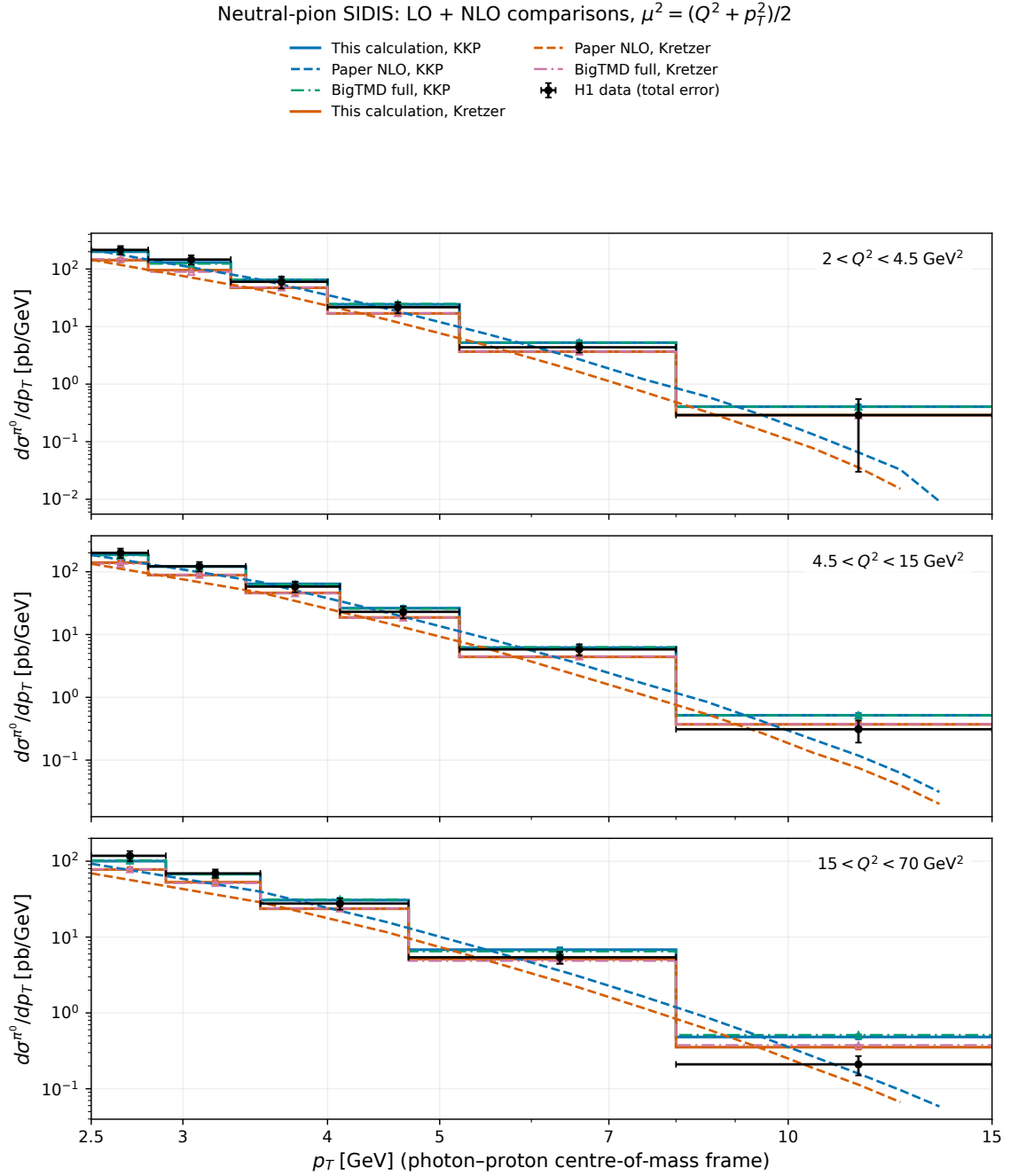
The hard-coefficient BigTMD comparison in `bigTMD_comparison/` and the plotted cross-section overlay answer different questions. The former compares partonic coefficient formulas under matched symbolic conventions. The latter includes PDFs, FFs, phase-space integration, cuts and numerical errors. The saved exact coefficient agreement is not a claim that independent Monte Carlo central values must coincide digit for digit.

19.3 Ratio definition and uncertainties

The source defines the plotted theory-to-data ratio and its integration error by

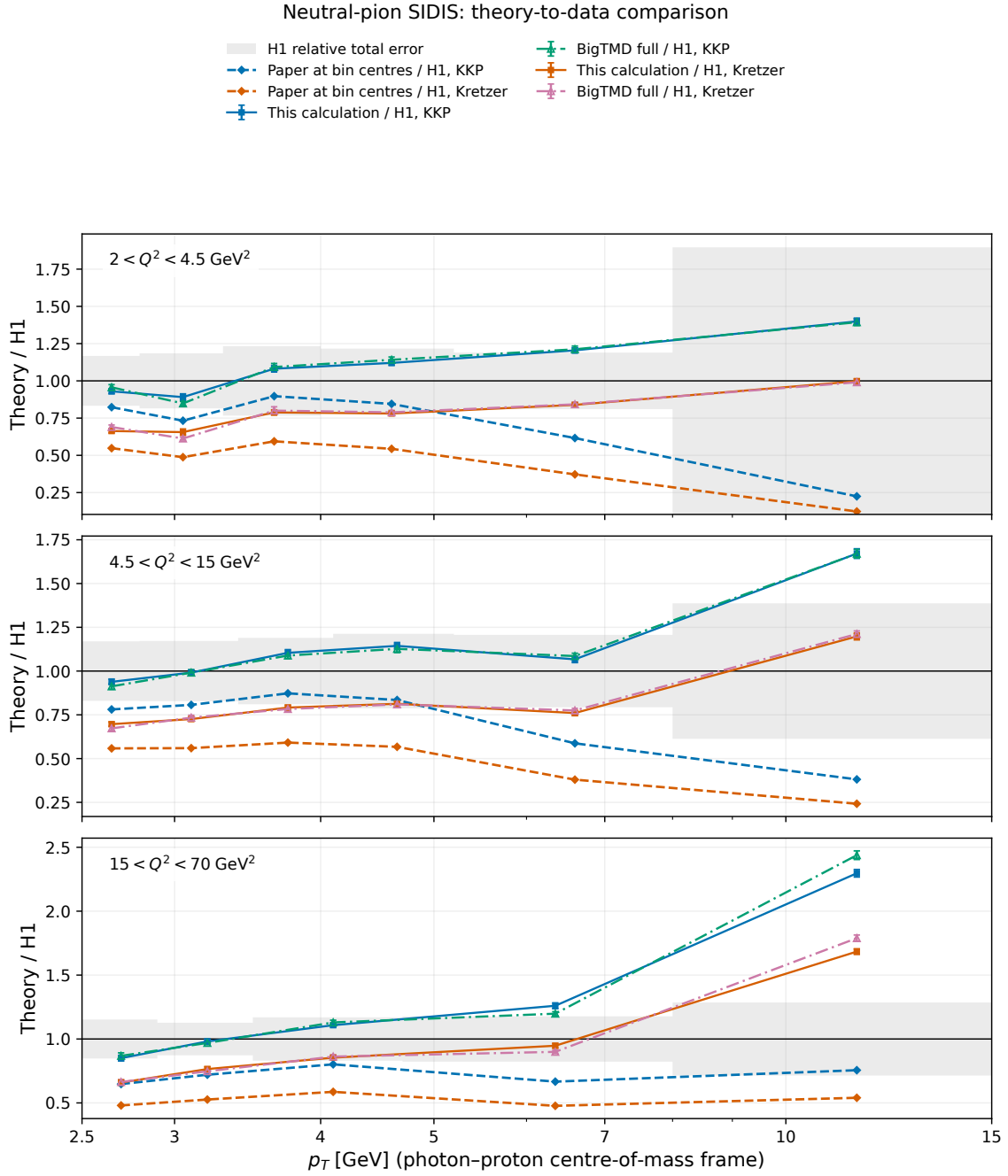
$$r_b = \frac{\langle d\sigma/dp_T \rangle_b^{\text{theory}}}{(d\sigma/dp_T)_b^{\text{H1}}}, \quad \delta r_b^{\text{int}} = \frac{\delta \langle d\sigma/dp_T \rangle_b^{\text{int}}}{(d\sigma/dp_T)_b^{\text{H1}}}. \quad (118)$$

The gray band in Figure 2 is the H1 relative total uncertainty around one. It is shown separately from the theory integration errors. The plots do not include PDF uncertainty, FF uncertainty, scale variation or an uncertainty from the azimuth approximation. The inherited Hgq MadGraph comparison remains a stated input caveat, as described in Section 9.



MRST2002 NLO; shared FFs, scales and H1 forward-pion cuts. Theory errors: integration only.
Hgg input; unresolved MadGraph real comparison retained; uniform azimuth.
BigTMD curves use complete coefficient formulas with the labelled channel selections.
Paper curves: original EPS vertices, hep-ph/0411212 Fig. 4 / 1808.04396 Fig. 3.

Figure 1: The saved neutral-pion LO+NLO transverse-momentum comparison using the six new SIDIS channels. Panels show the three Q^2 ranges. SIDIS and full BigTMD predictions are bin averages; paper inputs are curves. Theory errors represent numerical integration only. The setup and inherited Hgg caveat are stated in the text.



Our calculation and BigTMD use bin averages; paper values are interpolated at bin centres.
 Paper-centre ratios are curve-to-bin comparisons, not paper bin-integrated predictions.
 Gray band: experimental total error. Theory error bars: numerical integration only.

Figure 2: The saved theory-to-H1 ratios with KKP and Kretzer FFs. SIDIS and full BigTMD entries use bin averages; paper ratios use interpolated bin-centre values. The gray band shows experimental relative total errors, while theory bars show integration errors. Ratios use the definition in Eq. (118).

20 Files, reproducibility and the boundary of this report

20.1 Symbolic channel and shared artifacts

Each channel directory contains its sequential stage sources and the corresponding `sNN_result` folders. Its README records the channel conventions, normalization, accepted artifact dependencies and consumer boundaries. The `common/` directory holds implementations and results shared by channels, including Kira reduction databases, SubTropica inputs, evaluated master series and the final acceptance receipt. The layout adapter maps the original execution paths to the organized local folders; its preserved source and relocation records identify that path conversion.

The `.cluster/` directory contains the cluster execution support. The accepted production uses bounded workers and resumable caches, keeping the main source and result artifacts locally. Sharing one master reduction across the channels saves repeated integration work without merging their charge or spectator conventions. The current document expansion does not rerun those physics stages.

20.2 Numerical artifact sequence

The numerical README specifies the stage contracts. The sequence starts with frozen hard-function inputs and exports their branch and charge components. It then derives the physical kinematic maps, builds the native multiprecision evaluator, validates its PDF/FF interfaces and coefficient values, assembles flavor luminosities and integrates bins. The independent precision comparison precedes publication of the accepted plot bundle. The following records connect the equations and plots in this report to the retained outputs:

Path within <code>numerics/</code>	Content used here
<code>s04_result</code>	Defining physics inputs and symbolic maps, Jacobians, lepton weights and laboratory momentum.
<code>s06_result</code>	Numerical PDF, FF and coupling conventions and interface checks.
<code>s08_result</code>	Flavor, charge, neutral-FF and acceptance contracts used in the convolution.
<code>s09_result</code>	Seventeen accepted bins, components, covariance, iteration records and integrated cross sections.
<code>s10_result</code>	Independent coefficient precision checks.
<code>s11_result</code>	Accepted numerical aggregate and its output identity.
<code>dsigmatpbydpt/s03_result</code>	Bin averages, ratios, comparison inputs and figure identities.

The accepted numerical run records no OOM and a peak memory of about 1.48 GiB. Reusing its finished figures in this report requires only PDF typesetting. The source and regenerated report PDF are stored together in the SIDIS directory.

20.3 What an extension would have to supply

This is an NLO result. Reverse unitarity, Kira and explicit master evaluation provide a reusable integration organization, but this run does not establish an NNLO result. An extension would need its own higher-order amplitudes, cut and virtual families, master integrals and regulator depths, renormalization and factorization inputs, and final cancellation checks. None of those missing higher-order objects is supplied merely by the availability of the current two parent templates or by the agreement of the NLO coefficients with a reference.

References

- [1] B. Wang, J. O. Gonzalez-Hernandez, T. C. Rogers and N. Sato, *Large Transverse Momentum in Semi-Inclusive Deeply Inelastic Scattering Beyond Lowest Order*, arXiv:1903.01529v1 (2019). Supplied local paper in `scripts/`; especially Eqs. (9), (12), (16)–(19), (21)–(24), (32), (34), (38) and (46)–(53).
- [2] Supplied `standalone_nlo_qcd_reference/standalone_nlo_qcd-main/`, including `manual/standalone.tex` and its FeynCalc workflow. This supplies the reverse-unitarity/Kira/SubTropica integration approach. The actual SIDIS routing, physical cut representations and normalization are defined by the SIDIS programs.
- [3] Somogyi, arXiv:1101.3557, Eqs. (7), (56) and Appendix B. This is the unevaluated Euler-representation reference recorded in `common/s08_cut_master_inputs.wl` and its accepted result. SubTropica evaluates the SIDIS inputs built from that representation.
- [4] Local implementation and accepted results: `common/s02_definitions.wl`, `common/s04_reduce_real.wl`, `common/s08_cut_master_inputs.wl`, `common/s10_evaluate_cut_masters.wl`, `common/s14_uv_residues.wl`, `common/s17_assemble_real.wl`, `common/s19_assemble_virtual.wl` and `common/s20_final_hats.wl`; the six channel README.md files; and the corresponding numbered result directories. Original executed sources and the layout manifest are retained under `common/previous_runs/layout_sources/` and `common/s21_result/layout.json`.
- [5] Local numerical implementation and accepted outputs: `numerics/README.md`; `numerics/s04_derive_convolution_maps.py`, `numerics/s06_prepare_pdf_ff.py`, `numerics/s08_derive_consumer_actions.py`, `numerics/s09_convolve_channels.py`, `numerics/s10_validate_precision.py`, `numerics/s11_aggregate_replacement.py`; their numbered result files; and `numerics/dsigmapibydpt/s03_result`, which records the H1 points, paper-curve inputs, full BigTMD overlay, ratios and figure identities used in this report.